

TOPIC 12/60

What is a Context Window?

The AI's short-term memory.

Why ChatGPT sometimes forgets your name.

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THE READING LIMIT

An LLM can only process a specific amount of text at one time. This strict limit is called the **Context Window**.

The Golden Rule:

If a piece of information falls *outside* the context window, it physically does not exist to the AI.

> It cannot reference what it cannot see.

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THE ANALOGY



Goldfish Memory

Imagine having a conversation with a friend who can only remember the last 1,000 words spoken.

As you keep talking (adding new words), the oldest words get pushed out of their brain to make room. If you ask them about a joke you made 2 hours ago, they will hallucinate an answer because the joke was erased.

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THE MEASUREMENT

Measured in Tokens

Context windows are not measured in pages or characters; they are measured in **Tokens** (word chunks).

Standard Sizes

- **4K Tokens:** ~3,000 words (Early ChatGPT)
- **32K Tokens:** ~24,000 words (A short novel)
- **128K Tokens:** ~96,000 words (A Harry Potter book)

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THE HIDDEN CATCH

The Shared Budget

Input + Output

The context window is a single, shared budget for **both** what you type and what the AI generates.

If you have a 4,000 token window, and you paste a 3,800 token article into the prompt... the AI only has 200 tokens left to generate its summary before it crashes.

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THE FLAW

Lost in the Middle

Just because an AI has a massive 128k context window doesn't mean it reads perfectly.

The "U-Shape" Curve:

Research shows that when LLMs read huge documents, they perfectly remember the very beginning and the very end.

> But they often skip or hallucinate facts buried in the middle of the text.

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WHY ISN'T IT INFINITE?

QUADRATIC SCALING

The cost explodes exponentially.

Because of Self-Attention, every token must compare itself to every other token. Doubling the context window takes 4x the computing power and RAM.

THE HARDWARE LIMIT

GPU VRAM is expensive.

Processing 1 million tokens at once requires incredibly expensive, specialized server clusters.

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THE BREAKTHROUGH

The Million Era

Through clever engineering (like Ring Attention and FlashAttention), context windows are skyrocketing.

Google's Gemini 1.5 Pro stunned the world by launching a **2,000,000 token** context window. It can ingest 2-hour long videos, thousands of codebase files, or 50 books in a single prompt.

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THE PARADIGM SHIFT

Zero-Shot Context

You don't need to fine-tune a model on your company data. Just paste your entire 500-page manual into the prompt.

Complex Reasoning

The AI can find connections between page 3 and page 940 of a legal contract instantly.

The RAG Killer?

If models can hold 10 million tokens natively, complex RAG architectures (retrieving chunks of text) might become obsolete for many use cases.

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Context Window

CONCEPT UNLOCKED



The AI's short-term memory



Measured in Tokens



Prone to "Lost in the Middle"



Expanding to Millions of tokens

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